Original Article

A randomized controlled study evaluating outcomes of intense pulsed light and low-level light therapy for treating meibomian gland dysfunction and evaporative dry eye

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Purpose: This randomized, controlled, blinded study evaluates the efficacy of intense pulsed light (IPL) therapy with low-level light therapy (LLLT) in the treatment of meibomian gland dysfunction (MGD) and evaporative dry eye (EDE) compared to a control group. **Methods:** Hundred patients with MGD and EDE were randomized into control (50 subjects, 100 eyes) and study group (50 subjects, 100 eyes). The study group underwent three sittings of IPL with LLLT 15 days apart and were followed up 1 month and 2 months after the last treatment sitting. The control group underwent sham treatment and was followed up at the same intervals. The patients were evaluated at baseline and 1 month and 3 months (post 1st treatment) for dry eye. Schirmer's test and tear breakup time (TBUT), OSDI, meibomian gland expression, and meibography. **Results:** The study group showed significant improvement in OSDI scores ($P < 0.0001$) compared to the control group and a significant improvement in TBUT ($P < 0.005$) compared to the control group. There was no change in schirmer's test and an improvement in the meibomian gland expression but not significant. **Conclusion:** The results show that a combined therapy of IPL with LLT is effective in treating MGD with EDE compared to controls, and repeated treatment sessions have a cumulative effect on the disease outcomes.

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Dry eye disease (DED) is one of the most common and fastest-growing conditions prompting patients to visit ophthalmologists and to seek treatment. This places a substantial economic burden on both the society and patients and has a negative impact on the quality of life.[1] Meibomian gland dysfunction (MGD) is now considered a major cause of dry eye disease affecting as many as 70% of the population in some regions globally.[2,3] In this condition, the meibomian glands have structural and physiological changes resulting in reduced quantity and quality of meibum secretions.[4] Due to the stagnation of meibum, there is increased proliferation of bacteria which can change the lipid composition by secreted lipases and further reduce the gland expressibility.[5] In addition, there is an inflammatory component underlying DED which is related to the altered ocular surface homeostasis. Treatment of any type of DED is therefore aimed at treating the underlying condition and reducing the inflammation.[6,7] The traditional step-wise treatment for MGD includes warm compresses, eyelid hygiene, anti-inflammatory agents, macrolide antibiotics, and omega 3 fatty acids. All these have been known to provide relief in a substantial section of patients suffering from MGD associated dry eye disease; however, there are still many who are recalcitrant to therapy.[7-9] Therefore, newer therapeutic interventions for MGD have gained increasing popularity.[8,10,11]

Intense pulsed light (IPL) therapy for treatment of MGD is one such treatment option. The potential use of IPL in MGD was discovered in patients receiving IPL therapy for acne rosacea, who were also found to show improvement in their MGD status.[12] IPL in conjunction with low-level light therapy (LLLT) has today evolved from being an experimental modality for treatment of MGD to being proven to have additional beneficial effects in modulation of inflammatory markers, treating demodicosis and accelerate wound healing after lid surgeries.[13-15] The current study evaluates the effect of IPL with LLLT on MGD and evaporative dry eye disease.

Methods

This was a prospective, longitudinal, randomized, double-blinded study approved by the Institutional Ethics Committee. Subject recruitment and sample collection procedures were conducted as per institutional guidelines and in accordance with the tenets of the Declaration of Helsinki. IPL + LLLT was administered during three separate treatment sessions on Day 1, Day 15, and Day 30 as per manufacturer recommendations.

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A total of 100 subjects (200 eyes) with chronic MGD were included in the study after obtaining written informed consent, of which 50 subjects (100 eyes) were randomized to the study group and 50 patients (100 eyes) to the control group. Patients fulfilling the inclusion and exclusion criteria were randomized to study or control group. The researcher collecting clinical data was masked as to which eye was treated and participant was masked by not informing them as to which group they have been assigned to.

Each study and control subject underwent the same set of tests at baseline and 1 and 2 months after completing three sessions of therapy. The clinical evaluation of MGD and DED included Ocular Surface Disease Index (OSDI), tear breakup time (TBUT), schirmer's test 1 (ST1), corneal and conjunctival staining, lid evaluation including changes such as pitting, telangiectatic vessels, evaluation and grading of MGD based on the meibomian gland expressibility score (MGS), and meibography for meibomian gland dropout assessment. The patients also underwent a dilated retinal evaluation pre- and postprocedure. The schirmer's test 1 was performed for 5 minutes without topical anesthesia, using a sterile schirmer's test strip. Corneal and conjunctival staining was evaluated after instillation of fluorescein dye by wetting the strip with distilled water and instilling the dye using hanging drop method. TBUT was measured three times consecutively after fluorescein instillation and evaluating under blue filter on a slit lamp biomicroscope using a timer, and the median value was recorded. The OSDI questionnaire assessed the subjects frequency of dry eye symptoms and problems faced under tasks of daily living and environmental conditions over the previous week on a scale from 0 (none of the time) to 4 (all of the time).[16] Meibomian gland expressibility score (MGS) was assessed by expressing meibum from five glands each in the nasal, middle, and temporal parts of the eyelid. Secretions were graded and scored on a scale of 0–3 for each gland; 0, blocked; 1, toothpaste-like secretions; 2, cloudy secretions; and 3, clear secretions. The total score was calculated based on the number of glands multiplied by the type of meibum expressed. Maximum total score per eyelid was 45 and indicated normal meibomian glands.[17] Meibomian gland dropout on the eyelids was assessed and graded based on the percentage of dropout on meibography.[18] The Wong–Baker pain scale was measured immediately after each treatment to assess patient discomfort during the procedure.

Inclusion criteria

1) MGD grade 2 and above[19]; 2) chronic MGD of more than 6 months; 3) significant ocular discomfort while performing daily activities as measured by OSDI >23; 4) features of MGD based on clinical signs and meibography between 20% and 80% dropout; and 5) MGS (meibomian gland expressibility score) <12; 6) evaporative dry eye based on low TBUT <7 seconds and the Schirmer test >10 mm at 5 minutes; and 7) Fitzpatrick skin grading 1–4 were included in this study.

Exclusion criteria

Known allergy to the treatment; skin photosensitivity; active ocular infection; recent use of contact lenses <1 month back; recent ocular surgery (less than 6 weeks before); collagen vascular disease; history of procedural therapy for dry eye including punctum plugs in the eye; history of skin cancer; pigmentation in the proposed area of treatment; pregnant or lactating patients; and those with dark pigmented skin (Fitzpatrick grade >5) were excluded from treatment.[20] All subjects were monitored for treatment-related side effects. The plan of treatment and energy used for the IPL therapy are based on the severity of meibomian gland involvement on meibography and the skin tone graded by Fitzpatrick grading.[21]

Treatment details

The combined light therapy (IPL + LLLT therapy) was performed using the eye-light device (Espansione Marketing SPA., Bologna, Italy), which is CE marked for the treatment of MGD. Strict adherence to the safety protocols was observed while performing the procedure as per manufacturer's recommendations and existing guidelines.[22] The procedural treatment is administered after calculating the energy as per MGD and the Fitzpatrick skin-type grading. The study subjects underwent IPL, followed by LLLT as per the planned treatment and published recommendations.[22] As per safety guidelines, protective opaque goggles were used for the patient and their eyes kept closed while performing the procedure. Five IPL pulses are delivered around each eye at predetermined standard sites. During the LLLT, the patient was advised to keep the eyes closed.

The sham IPL treatment for the control group was simulated by placing the device over the same five spot zones around each eye, but no flash was released. For the LLLT, the mask was placed over the face of control subjects for 15 minutes as in study subjects, but treatment not started.

For both groups, the procedure was performed by the same trained ophthalmologist, and three sittings of the treatment were done at 0, 15, and 30 days, and results were assessed at 1 and 2 months postprocedure (3 months post first treatment). To avoid confounding factors influencing the results of therapy, no mechanical or manual expression of meibomian glands was performed for these patients during the study period. However, they were allowed to continue topical therapy they were using prior to enrolment in the study.

Statistical analysis

Results of tests done after three sessions of IPL + LLLT were compared to the baseline results of the same tests in the subjects and control groups. Data was compared across different time points, and paired analysis was carried out between the pre- and post-treatment data.

Observations are reported as mean +/- standard error of mean and represented as bar graphs. Differences in the variables between the pretreatment and post-treatment in matched samples were analyzed using the paired *t*-test using MedCalc v19 (MedCalc Inc, Belgium). $P < 0.05$ was considered to be statistically significant.

Results

The age of patients varied between 22 and 78 years with average age being approximately 48 years. The baseline parameters were comparable between the study and control group across age, gender, clinical evaluation and Fitzpatrick skin pigment grade [Table 1]. There were no patients in the Fitzpatrick skin pigment grades 1 and 2 in the study or control group.

Table 1: Baseline characteristics across the study group and the control group

Baseline characteristics (Mean±SD)	Study	Control
AGE	47.92±16.56	49.76±16.19
M/F	20/30	24/26
FITZPATRICK GRADE	3.08±0.71	3.14±1.11
SCHIRMERS	19.89±7.04	18.85±5.38
TBUT	5.19±2.75	5.09±1.79
OSDI	37.88±16.02	35.84±16.99
MEIBOGRAPHY	30.45±7.99	32.81±10.24

Clinical parameters

Patients who had under gone IPL + LLLT showed a significant improvement in clinical metrics compared to the control group at both postprocedure visits. There was no significant change noted in Schirmer value pre- and post-treatment in the study or the control group [Fig. 1a]. The OSDI score was observed to be significantly ($P < 0.0001$) reduced 1 month post-treatment and 3 months post-treatment compared with pretreatment as shown in Fig. 1b. However, no significant change was seen in the control group across the same time period even though there was an improvement. A significant increase ($P < 0.005$) in the tear breakup time (TBUT) was observed 1 month post-treatment and 3 months post-treatment compared with pretreatment in the study group [Fig. 1c]. The control group did show an improvement, but the extent of improvement was much more in the study group. There seems to be a cumulative effect of the treatment over time, with a progressive improvement seen in both the OSDI and TBUT scores over two visits [Fig. 1b and c].

The MGS showed an improvement post the IPL + LLLT; however, this was not statistically significant and was comparable to the control group. There was also an improvement in the ocular surface staining and reduction in visible congestion in more than 50% of the patients in the study group. Patients were very comfortable during the procedure with the maximum score on the Wong–Baker pain scale being 2. The findings strongly indicate that combined light therapy improves ocular surface-related signs and symptoms in the study subjects. There were no side effects or complications of the procedure noted during the study period.

Discussion

Meibomian gland disease can result in changes in the tear film stability and evaporative dry eye.[19] This study demonstrates the clinical efficacy of IPL with LLLT in treating MGD as shown in previous studies.[8,13,22,23] Significant improvement in symptomatology along with improvement in tear film parameters like tear breakup time was demonstrated. There was an improvement in ocular symptoms as demonstrated by significant reduction in OSDI in the study group after three treatment sessions. There are few randomized control, masked trials which have studied the effect of IPL on MGD and evaporative dry eye.[21,24,25] The significantly improved tear film stability in this study ($P < 0.005$) suggests that IPL + LLLT has an effect on the meibomian gland outflow both in the quantity and quality of meibum secreted. The results suggest a cumulative benefit of applying three

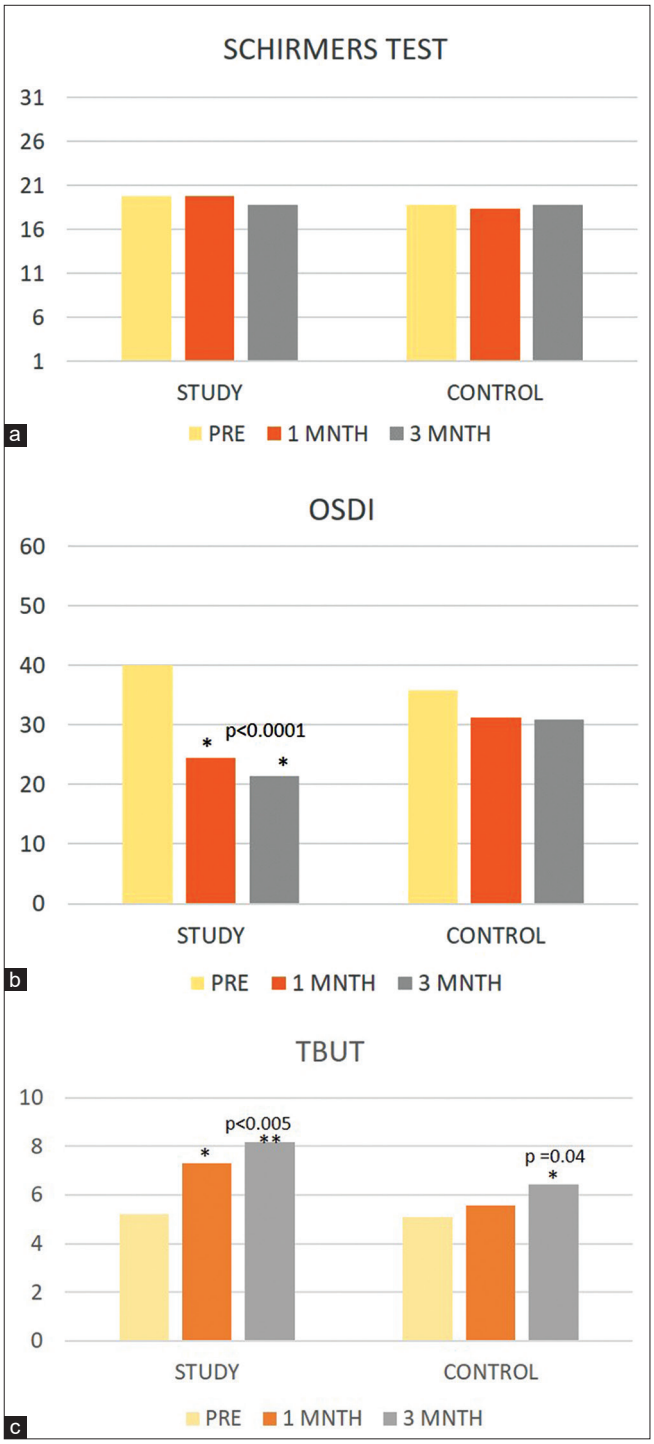


Figure 1: (a) Shows the average value of Schirmer test in the pretreatment (PRE), 1 month post-treatment (1 MNTH) and 3 months post-treatment (3 MNTH). No significant change was noted compared to the pretreatment values in the study or control group. (b) Shows the OSDI values in the pretreatment (PRE), 1 month post-treatment (1 MNTH) and 3 months post-treatment (3 MNTH). Significant improvement was noted from pretreatment to post-treatment in the study group, but no significant change was noted in the control group. (c) Shows the TBUT values in the pretreatment (PRE), 1 month post-treatment (1 MNTH) and 3 months post-treatment (3 MNTH). Significant improvement was noted in the study as well as control group, but the study group showed a much higher improvement than the control group

treatments with significant improvement of tear breakup time and discomfort score over a period of 3 months in the study subjects as compared to controls. This correlates with results from previous studies which hypothesized that a single treatment session could give a relatively shorter term of improvement as compared to the cumulative effect of three treatments.[8]

This improvement in tear breakup time is similar to results seen in previous studies.[8,26-28] The additional improvement in OSDI in the study subjects compared to controls ($P < 0.0001$) suggests that the clinical signs and symptoms are improved by the procedure and could be influenced by the multimodal effects of therapy including its effect on ocular surface inflammation. Other features like meibomian gland expressibility and telangiectasia also showed an improvement; however, as meibomian gland expression was not carried out in these patients to avoid confounding the results, this improvement is mainly related to improved quality of the secretions.

There are various mechanisms by which IPL has been proposed to work including reducing inflammation and promoting an antinociceptive molecular profile on the ocular surface. This could explain the long-term benefits of the procedure and the improved symptomatology.[8] The cumulative effect could also be due to the thrombosis of microvasculature around the meibomian glands and less inflammatory mediator release.[29] LLLT is purported to act by photobiomodulation using a near-infrared wavelength light.[22] It has been shown to help in wound healing and normal cell function.[30,31] LLLT can have an effect on the meibomian glands in the upper lid since IPL is only performed close to the lower eyelids.

The short follow-up of patients postprocedure is a limiting factor of the study as we expect the effect of IPL to last for longer and have a cumulative effect. However, since there was a control group who could potentially benefit from the therapy, both groups were followed by for an equal time period and then reassessed to see if any additional therapy was required as ethically necessitated. Another limiting factor was the continuation of topical medications the patient was on, even after the procedure. This was mainly done to ensure that the control group does not need to stop their medications to maintain comparability.

Conclusion

The intense pulse light therapy with low-level light therapy has shown good results for the treatment of MGD and evaporative dry eye. Cumulative effects of sequential sessions have been evaluated and found to have good results in this study. In addition, the presence of the control arm clearly further highlights the effectiveness of this therapy. No adverse effects like postprocedure skin pigmentation or burns were noted during this study. Long-term results and evaluation of efficacy across additional diagnoses like mixed dry eye need to be evaluated.

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Conflicts of interest

There are no conflicts of interest.

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